Shared-leg rectangular matrix multiplication over arbitrary rings

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Abstract

We give an integral extension of the rectangular bound in arXiv:2610.10494v1. The new algebraic ingredients are an integer asymptotic-rank bound for convolution and an integer replacement for Fourier separation. Together with a discrete proof of the real-variable profile implication, they produce integer bilinear algorithms valid over every associative ring, including noncommutative rings and rings without identity. All bounds concern arithmetic operations, with arbitrary positive slack in the exponent; no bit-complexity bound is asserted.

1 Statement and computational model

For $k \geq 0$ set

$$f(k) = \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \dfrac{1}{4k}, & k > \frac{1}{2}. \end{cases}$$

Theorem 1.1 (Universal ring version)**.** *For every $k \geq 0$ and $\varepsilon > 0$ there are a family of integer bilinear circuits and a constant $C = C(k, \varepsilon) > 0$ such that, for every associative ring R and every integer $n \geq 1$, the circuit for size n multiplies an $n \times \lceil n^k \rceil$ matrix by a $\lceil n^k \rceil \times n$ matrix using at most $Cn^{f(k)+\varepsilon}$ additions, negations, and multiplications. The circuits and C are chosen before R. Commutativity, an identity, and restrictions on characteristic or zero divisors are unnecessary.*

Here an integer bilinear algorithm forms integer linear combinations of the first input, integer linear combinations of the second input, multiplies each first combination on the left of its second combination, and recombines the products with integer coefficients. Integer multiples are implemented by additions and negations. For $c \in \mathbb{Z}$ and $r \in R$, cr means the integer action on the additive group: repeated addition, with negation for negative c. It requires neither an element c in R nor a multiplicative identity. Each addition, negation, or ordered multiplication costs one operation. Input reads, zero, copying and reindexing available values, and choosing output coordinates cost no arithmetic operations. Finite intermediate families are computed once and may be reused. There is no primitive integer scalar multiplication: it is compiled into additions and negations. The positive exponent slack absorbs logarithmic overhead; the theorem does not assert $O(n^{f(k)})$ without slack.

Background used. For background on bilinear algorithms and block recursion, consult [1, Chapters 14–15]; for finite fields, [2, Chapter 2]. We use elementary finite free module and determinant algebra, the existence of finite fields of every prime-power order, and standard real analysis (including Fekete's lemma and Stirling's estimate). Its only specialized external theorem is abstract spectral rank duality, stated below with its hypotheses. The integral constructions and discrete parameter argument are proved here; spectral duality itself is cited rather than reproved. A declaration-level correspondence with the checked Lean development is in `formalization/proof-correspondence.md`.

Proof route. An integer coefficient identity gives an algorithm over every ring because its two multiplication factors retain their order. It therefore suffices to obtain one fixed integer base algorithm with the required rank-to-size ratio. Integral spectral duality reduces that task to bounding character values of matrix tensors. Coefficient extraction and convolution supply the shared-leg entropy inequalities; the discrete profile argument then bounds the character parameters by a parabola. The final step selects the base algorithm and charges all coefficient arithmetic in its recursion.

2 An integral spectrum with the right order

An integer tensor is a finite three-index array T_{ijh}, written as the formal expression

$$T(x,y,z) = \sum_{i,j,h} T_{ijh} x_i y_j z_h.$$

The three variable groups are its *legs*. These variables record coefficients; they do not impose commutativity on the ring where the algorithm will run. A rank-one tensor is $L(x)M(y)N(z)$, with integer linear forms. Thus $T = \sum_{\ell=1}^{r} L_\ell(x)M_\ell(y)N_\ell(z)$ describes r ordered products: evaluate L_ℓ on the first input and M_ℓ on the second, multiply in that order, and use the coefficients of N_ℓ to recombine the outputs. Tensor rank counts these products; the final section also charges additions and negations.

A restriction independently substitutes integer linear forms on each leg. Direct sum uses disjoint coordinate groups on all three legs, with every mixed coefficient zero. Tensor product pairs coordinate indices and multiplies the corresponding integer coefficients. All coordinate spaces below are finite free $\mathbb{Z}$-modules. Write $A \geq_{\mathbb{Z}} B$ when A restricts to B. For $m \in \mathbb{N}$, let $D_m = \sum_{i=1}^{m} x_i y_i z_i$ and $1 = D_1$. The integer rank $\operatorname{rank}_{\mathbb{Z}}(T)$ is the least m with $D_m \geq_{\mathbb{Z}} T$.

We work in the semiring $\mathcal{S}$ consisting of 0 and the integer tensors that restrict to 1, modulo mutual integer restriction. Addition is full direct sum and multiplication is tensor product. This choice matters: all nonzero integer tensors would be unsuitable, since $2xyz$ cannot restrict to xyz over $\mathbb{Z}$.

The semiring is closed under both operations and under leg permutations. Every nonzero $T \in \mathcal{S}$ satisfies

$$D_{\operatorname{rank}_{\mathbb{Z}}(T)} \geq_{\mathbb{Z}} T \geq_{\mathbb{Z}} 1.$$

Expansion in a basis proves finiteness of rank. The diagonal tensors embed $\mathbb{N}$ with its usual order: the reverse implication follows by flattening after base change to $\mathbb{Q}$. Restrictions respect both operations. Finally, for any A and nonzero B, some m satisfies $mB \geq_{\mathbb{Z}} A$, because $B \geq_{\mathbb{Z}} 1$ and $D_{\operatorname{rank}_{\mathbb{Z}}(A)} \geq_{\mathbb{Z}} A$. Thus restriction is a Strassen preorder on $\mathcal{S}$.

Here the preorder convention for the abstract theorem is $B \preceq A$ if $A \geq_{\mathbb{Z}} B$. The theorem applies to a commutative semiring with a preorder that induces the usual order on $\mathbb{N}$, respects addition

and multiplication, and has the cofinality property $A \preceq mB$ for some $m \in \mathbb{N}$ whenever $B \neq 0$. It asserts that the normalized monotone semiring homomorphisms form a nonempty spectrum and, when $1 \preceq T$, their supremum is the asymptotic rank of T. The preceding paragraph verifies each hypothesis; commutativity here concerns tensor classes, independently of the eventual coefficient ring of the algorithm.

Let $\Delta(\mathcal{S})$ be the set of normalized, additive, multiplicative, restriction-monotone maps $\lambda : \mathcal{S} \to \mathbb{R}_{\geq 0}$. Set $\widetilde{\mathrm{rank}}_{\mathbb{Z}}(0) = 0$ and, for nonzero $T \in \mathcal{S}$, define

$$\widetilde{\mathrm{rank}}_{\mathbb{Z}}(T) = \exp\left(\inf_{m \geq 1} \frac{\log \mathrm{rank}_{\mathbb{Z}}(T^{\otimes m})}{m}\right).$$

Rank is submultiplicative under tensor product, so the logarithms are subadditive. Fekete's lemma identifies this infimum with the limit in the next display. This infimum definition is the one used in Lean. The abstract spectral rank theorem [5, Theorem 2.13, Corollary 2.14, Theorem 2.16] therefore gives, for nonzero $T \in \mathcal{S}$,

$$\widetilde{\mathrm{rank}}_{\mathbb{Z}}(T) = \lim_{m \to \infty} \mathrm{rank}_{\mathbb{Z}}(T^{\otimes m})^{1/m} = \sup_{\lambda \in \Delta(\mathcal{S})} \lambda(T). \tag{1}$$

The nonzero case has $T \geq_{\mathbb{Z}} 1$, so every character has value at least 1, as required by that theorem. For $T = 0$, both rank and every character value are zero, and the same display holds directly. We call these maps characters. In particular, $1 \leq \lambda(T) \leq \mathrm{rank}_{\mathbb{Z}}(T)$ for nonzero objects, and $\lambda(T) \leq \widetilde{\mathrm{rank}}_{\mathbb{Z}}(T)$ by applying the rank bound to tensor powers. Thus a character is a numerical size estimate compatible with independent tasks, tensor products, and integer preprocessing. Each character gives a rank lower bound; spectral duality says that bounding all of them also bounds asymptotic rank.

For a positive integer q, the oriented dot products are

$$B_X(q) = x \sum_{j=1}^{q} y_j z_j, \quad B_Y(q) = y \sum_{j=1}^{q} x_j z_j, \quad B_Z(q) = z \sum_{j=1}^{q} x_j y_j.$$

They belong to $\mathcal{S}$. For each character, multiplicativity and monotonicity give parameters $p_i = \log_2 \lambda(B_i(2))$, $i \in \{X, Y, Z\}$, with

$$\lambda(B_i(q)) = q^{p_i}, \qquad 0 \leq p_i \leq 1. \tag{2}$$

Comparison of q^m with adjacent powers of 2 proves the power law. For positive integers a, b, c, define the matrix multiplication tensor

$$\langle a, b, c \rangle = \sum_{i=1}^{a} \sum_{j=1}^{b} \sum_{h=1}^{c} x_{ij} y_{jh} z_{hi}.$$

Its factorization into the three oriented dot products gives

$$\lambda(\langle a, b, c \rangle) = a^{p_Y} b^{p_Z} c^{p_X}. \tag{3}$$

3 Coefficient extraction replaces division

Write $A \rightsquigarrow_{\mathbb{Z}} S$ when integer Laurent-polynomial leg maps transform A to a tensor family $F(t) = t^{\ell} S +$ terms of degree greater than ℓ for some $\ell \in \mathbb{Z}$. This also allows $S = 0$. In the weighted constructions below, a variable of integer weight w is replaced by t^w times that variable. The lowest total-weight terms give the leading coefficient. The parameter t is a formal bookkeeping device; coefficient extraction makes the resulting comparison integral.

Lemma 3.1 (Integral degeneration monotonicity)**.** *If $A, S \in \mathcal{S}$ and $A \rightsquigarrow_{\mathbb{Z}} S$, then $\lambda(A) \geq \lambda(S)$ for every character.*

Proof. Shift the leg maps so their entries are polynomials and the shifted ℓ is nonnegative. Write the resulting tensor as $F(t) = t^{\ell}S + \text{higher-degree terms}$, $\ell \geq 0$. For an integer $d \geq 0$, the coefficient tensor on three copies of $\mathbb{Z}^{d+1}$ is

$$H_d = \sum_{\substack{i,j,h \geq 0 \\ i+j+h=d}} u_i v_j w_h.$$

It has a coefficient 1 and at most $(d+1)^2$ terms, so it lies in $\mathcal{S}$ and has rank at most $(d+1)^2$. For each $m \geq 1$, extracting degree $m\ell$ from the powered leg maps gives the restriction

$$A^{\otimes m} \otimes H_{m\ell} \geq_{\mathbb{Z}} S^{\otimes m},$$

explicitly: if $L_X(t)$ is the powered first-leg map, send $x \otimes u_i$ to $[t^i]L_X(t)x$, and use the analogous maps on the other legs. The condition $i + j + h = m\ell$ extracts exactly that coefficient of $F(t)^{\otimes m}$, which is $S^{\otimes m}$. Thus

$$\lambda(S)^m \leq (m\ell + 1)^2 \lambda(A)^m.$$

The extra factor tends to 1 after taking mth roots. Here ℓ is fixed by the original degeneration, independently of m. □

4 The integral convolution bound

For positive r, s define

$$C(r,s) = \sum_{i=0}^{r-1} \sum_{j=0}^{s-1} x_i y_j z_{i+j}, \qquad N = r + s - 1.$$

Its coefficient of $x_0 y_0 z_0$ is 1, so $C(r,s) \in \mathcal{S}$. The later separator needs only $\lambda(C(r,s)) \leq N$. A single scaled interpolation identity cannot give this over the integers: its multiplier must also be eliminated. We construct finitely many identities whose multipliers can be patched by Bézout.

Lemma 4.1. *For every fixed $r, s \geq 1$, with $N = r + s - 1$, there is a constant $K_{r,s} > 0$ such that*

$$\operatorname{rank}_{\mathbb{Z}}(C(r,s)^{\otimes m}) \leq K_{r,s} N^m \quad (m \geq 1).$$

Consequently every character satisfies $\lambda(C(r,s)) \leq N$.

Proof. Let $B = \mathbb{Z}[t]/(g)$, where g is monic of degree $d \geq 1$; $B = \mathbb{Z}$ with $d = 1$ is also allowed. Its standard basis starts with 1. For any nodes $a_0, \ldots, a_{N-1} \in B$, form the Vandermonde matrix $V_{\ell k} = a_\ell^k$ $(0 \leq \ell, k < N)$, and put $\delta = \det V$. The identity $\operatorname{adj}(V)V = \delta I$ gives

$$\delta C(r,s) = \sum_{\ell=0}^{N-1} \left(\sum_{i=0}^{r-1} a_\ell^i x_i \right) \left(\sum_{j=0}^{s-1} a_\ell^j y_j \right) \left(\sum_{k=0}^{N-1} \operatorname{adj}(V)_{k\ell} z_k \right).$$

Define $E = \operatorname{Norm}_B(\delta) \in \mathbb{Z}$ as the determinant of multiplication by δ on the free integer module B. Its adjugate supplies $\gamma \in B$ with $\gamma\delta = E \cdot 1_B$: if M_δ is that multiplication matrix, take the coordinates of γ to be $\operatorname{adj}(M_\delta)$ times the coordinates of 1_B. The identity $M_\delta \operatorname{adj}(M_\delta) = EI$ proves

the claim. Multiplying the output forms above by γ therefore gives a length-N decomposition of $EC(r,s)$ over B. Both adjugate identities hold without inversion.

Take the mth tensor power *in this same algebra*. It decomposes $E^mC(r,s)^{\otimes m}$ with N^m terms. To descend it once to integers, let $\pi : B \to \mathbb{Z}$ extract the coefficient of 1. For $u, v, w \in B$, write $u = \sum_{a<d} u_a t^a$ and $v = \sum_{b<d} v_b t^b$. Then

$$\pi(uvw) = \sum_{0\le a,b<d} u_a v_b \, \pi(t^{a+b} w).$$

Since $\pi(1) = 1$, this produces an integer decomposition with at most d^2N^m terms. Only $\mathbb{Z}$-linearity of π is used. Descending after powering keeps the overhead d^2 independent of m.

First choose $B_0 = \mathbb{Z}$ and the nodes $0, 1, \dots, N-1$. The resulting Vandermonde determinant is a nonzero integer E_0. For each prime $p \mid E_0$, choose $d_p \ge 1$ with $p^{d_p} \ge N$ and a monic integer lift g_p of an irreducible polynomial of degree d_p over $\mathbb{F}_p$. The algebra $B_p = \mathbb{Z}[t]/(g_p)$ reduces to $\mathbb{F}_{p^{d_p}}$. Choose N nodes with distinct residues. The Vandermonde determinant is nonzero in that residue field; its norm, reduced modulo p, is therefore nonzero. Thus the integer multiplier E_p satisfies $p \nmid E_p$.

Let $I = \{0\} \sqcup \{p : p \text{ is prime and } p \mid E_0\}$ and $d_0 = 1$. The multipliers $(E_i)_{i\in I}$ have gcd 1, as do their mth powers. For each $m \ge 1$, choose integers $b_{i,m}$ such that

$$\sum_{i\in I} b_{i,m} E_i^m = 1.$$

Multiply the descended rank decompositions by $b_{i,m}$ in their third legs and add them. This gives an exact integer decomposition of $C(r,s)^{\otimes m}$ whose number of terms is at most

$$\left(\sum_{i\in I} d_i^2\right) N^m.$$

All algebras, nodes, and dimensions were fixed before m; hence $K_{r,s} = \sum_{i\in I} d_i^2$ is a fixed positive constant. Applying a character and taking mth roots gives the required bound $\lambda(C(r,s)) \le N$. □

Remark 4.2. The output flattening over $\mathbb{Q}$ has rank N, and its tensor powers have rank N^m, so the bound also gives $\widetilde{\operatorname{rank}}_{\mathbb{Z}}(C(r,s)) = N$. The auxiliary algebras appear only in the proof of an integer identity. For example, $C(2,2)$ uses the ordinary nodes $0, 1, 2$ with multiplier $E_0 = 2$, and the nodes $0, 1, t$ in $\mathbb{Z}[t]/(t^2+t+1)$ with multiplier $E_2 = 3$. Thus the powers of 2 and 3 can be combined by Bézout. The second multiplier removes the prime obstruction left by the first.

5 An integral shared-leg separator

Character additivity applies to full direct sums. Shared-X sectors must therefore acquire independent X labels. The convolution below tracks the difference of two sector labels, and a quadratic weight retains precisely the terms where they agree.

Proposition 5.1. *Let $M \ge 1$, and let X, Y_h, Z_h be finite free coordinate modules. Suppose $A = \sum_{h=1}^{M} A_h$, with $A_h \in X \otimes Y_h \otimes Z_h$, where the Y_h are disjoint, the Z_h are disjoint, and $A_h \in \mathcal{S}$. There is an integer restriction followed by an integral degeneration*

$$A \otimes C(2M-1, 2M-1) \rightsquigarrow_{\mathbb{Z}} \bigoplus_{h=1}^{M} \big(A_h \otimes B_X(M)\big).$$

The X-spaces on the right are independent copies.

Proof. Choose coordinate bases indexed by a in X, and by b, c in Y_h, Z_h. Let c_{habc} be the coefficient of $x_a y_{hb} z_{hc}$ in A. The convolution coordinates ξ_i, η_j, ζ_k have ranges $0 \le i, j \le 2M-2$ and $0 \le k \le 4M-4$. Introduce target variables X_{ag}, Y_{hbu}, Z_{hcv}, where $g, u, v \in \{1, \dots, M\}$. Use the following three substitutions, with every unmatched coordinate sent to zero:

$$x_a \xi_i \longmapsto \sum_{g:\ i=2g-2} X_{ag},$$
$$y_{hb} \eta_j \longmapsto \sum_{u:\ j=u-h+M-1} Y_{hbu},$$
$$z_{hc} \zeta_k \longmapsto \sum_{v:\ k=v+h+M-3} Z_{hcv}.$$

All indicated indices lie in the convolution ranges. The support equation $i + j = k$ becomes

$$u - v + 2(g - h) = 0.$$

Give the target variables the integer weights

$$w(X_{ag}) = g^2, \qquad w(Y_{hbu}) = hu - h^2, \qquad w(Z_{hcv}) = -hv.$$

On the supported terms their sum is $(g-h)^2$. The lowest-weight terms consequently have $g = h$ and $u = v$, and are exactly

$$\sum_{h,a,b,c,u} c_{habc} X_{ah} Y_{hbu} Z_{hcu}.$$

This is the stated full direct sum. Every substitution has integer coefficients; no roots of unity or normalization denominators occur. To obtain the nonnegative polynomial exponents used in Lean, replace the three weights by $w_X, w_Y + M^2, w_Z + M^2$. Their supported sum is $2M^2 + (g-h)^2$, so the same target is the coefficient of degree $2M^2$. □

Corollary 5.2 (Integral shared-leg entropy)**.** *Let $s \ge 1$ and $T = \sum_{i=1}^{s} T_i$ have the matched-sector structure of Proposition 5.1, with $T_i \in \mathcal{S} \setminus \{0\}$. For every character and every probability vector $q = (q_1, \dots, q_s)$, $q_i \ge 0$, $\sum_i q_i = 1$,*

$$\lambda(T) \ge \exp(p_X H(q)) \prod_i \lambda(T_i)^{q_i}, \qquad H(q) = -\sum_i q_i \log q_i,$$

where $0 \log 0 = 0$. For $A \in \mathcal{S}$ with $A \rightsquigarrow_{\mathbb{Z}} T$, the same lower bound holds with $\lambda(A)$ on the left.

Proof. Deleting every sector except one restricts T to a nonzero T_i, so T itself admits the unit. In a tensor power, the second and third legs retain their sector-word labels, whereas the first leg is shared. For rational q, let $m \to \infty$ through multiples of a common denominator. Retain in $T^{\otimes m}$ only matched words with mq_i occurrences of sector i, by deleting all other second- and third-leg coordinates. Different retained words have zero mixed coefficients, so they are matched sectors. There are $M = m! / \prod_i (mq_i)!$ sectors, each having character value $\prod_i \lambda(T_i)^{mq_i}$. Lemmas 3.1, 4.1 and Proposition 5.1 give

$$(4M-3)\lambda(T)^m \ge M^{1+p_X} \prod_i \lambda(T_i)^{mq_i}.$$

Taking mth roots leaves only

$$\lambda(T) \ge M^{p_X/m} \prod_i \lambda(T_i)^{q_i} \left(\frac{4M-3}{M}\right)^{-1/m}.$$

Since $1 \le (4M-3)/M \le 4$ and $\log M = mH(q) + O_s(\log(m+1))$, the result follows. The last estimate is Stirling's formula applied to the factorials, with $0! = 1$ for zero counts. Continuity extends it to all probability vectors. For each fixed m the degeneration has already been handled by Lemma 3.1; no uniform degree estimate in the type-counting limit is needed. □

6 Integer sector constructions

The entropy inequality now applies to two integer constructions: a filtration supplies concavity, and a three-sector degeneration couples the two profiles. The orientations used below are $C_{XY} = C$ and $C_{XZ}(r,s) = \sum_{i<r,j<s} x_i z_j y_{i+j}$, with nonnegative indices.

Determinant filtration. Fix positive integers r, s. Let $\mathcal{V}_e = \mathbb{Z}[u,v]_e$ be the free module of homogeneous binary forms of degree $e \ge 0$, and let σ, τ be a basis of an auxiliary rank-two module. In $\mathcal{V}_e \otimes \langle \sigma, \tau \rangle$, use

$$A_i = u^{e-i}v^i\sigma, \quad B_i = u^{e-i}v^i\tau \quad (0 \le i \le e), \qquad K_j = B_j - A_{j+1} \quad (0 \le j < e).$$

The basis $(A_0, \dots, A_e, B_e, K_0, \dots, K_{e-1})$ is unimodular: its inverse is $B_j = K_j + A_{j+1}$. Evaluation $\sigma \mapsto u$, $\tau \mapsto v$ sends the first $e+2$ vectors to the monomial basis of $\mathcal{V}_{e+1}$. For $e \ge 1$, the kernel is $(u\tau - v\sigma)\mathcal{V}_{e-1}$; for $e = 0$, it is zero.

In monomial bases, multiplication of forms with r and s coefficients is exactly $C(r,s)$: the output coefficient of index k is $\sum_{i+j=k} x_i y_j$. The auxiliary rank-two identity contributes the factor $B_X(2)$, leaving the coefficient leg unchanged. For $F \in \mathcal{V}_{r-1}$, multiplication by F respects evaluation and its kernel. At input degree $s-1$ and output degree $r+s-2$, its quotient and kernel blocks are multiplication

$$\mathcal{V}_s \longrightarrow \mathcal{V}_{r+s-1}, \qquad \mathcal{V}_{s-2} \longrightarrow \mathcal{V}_{r+s-3} \quad (s \ge 2).$$

These are $C_{XY}(r, s+1)$ and $C_{XY}(r, s-1)$, sharing the coefficient leg of F. Weight -1 on the input kernel and $+1$ on the output dual kernel keeps both diagonal blocks at weight zero. The block from input kernel to output quotient is zero, because multiplication preserves the kernel. The opposite off-diagonal block has weight one and disappears from the leading coefficient. Thus $C_{XY}(r,s) \otimes B_X(2)$ degenerates onto these two matched sectors, with a common first leg and disjoint second and third legs. At $s = 1$, the input kernel is empty, leaving only $C_{XY}(r,2)$. Permuting legs gives the other orientations.

Three sectors. In $C(r, 3s+r-1)$, the matched sectors have coordinates $(0 \le a < r,\ 0 \le b < s)$

	x	y	z
left	a	b	$a+b$
middle	$r-1-a$	$s+a+b$	$s+r-1+b$
right	a	$2s+r-1+b$	$2s+r-1+a+b.$

Give the middle Y-interval $[s, 2s+r-2]$ weight 1 and the middle Z-interval $[s+r-1, 2s+r-2]$ weight -1; all other weights are zero. The convolution support prevents negative-weight terms. The zero-weight tensor consists of the two outer $C_{XY}(r,s)$ sectors and the middle $C_{XZ}(r,s)$ sector, with the first coordinate reversed in the middle sector. That reversal identifies the value of the individual middle branch; the actual three branches continue to share the original first leg.

Character profiles. The profiles are numerical functions of the two convolution dimensions. A character may distinguish the first two legs. Pairing it with the exchanged character makes one profile symmetric while retaining a second profile for the different output orientation. For a character λ and its parameters from (2), define $G : \mathcal{S} \to \mathbb{R}_{\geq 0}$ and a_0, b_0 by

$$G(T) = \sqrt{\lambda(T)\lambda(T^{X\leftrightarrow Y})}, \qquad a_0 = (p_X + p_Y)/2, \quad b_0 = p_Z,$$

Here $T^{X\leftrightarrow Y}$ exchanges the first two legs. For $r, s \geq 1$, define the symmetric and oriented convolution profiles

$$V(r,s) = G(C_{XY}(r,s)), \qquad U(r,s) = G(C_{XZ}(r,s)).$$

The dot law, determinant endpoint, and output-dot restriction of $C(2,2)$ give the bounds used at zero parameters:

$$V(r,1) = r^{a_0}, \quad V(r,2) \leq 2^{a_0} V(r,1), \quad 2^{b_0} \leq V(2,2).$$

For later leg permutations, write $\lambda^{\pi}(T) = \lambda(T^{\pi})$, where XY, YZ exchange the named legs and $T^{\mathrm{cyc}}(y,z,x) = T(x,y,z)$. The following identities follow by relabeling the convolution indices. "Paired" means the geometric mean of the two character values:

characters	arguments	paired C_{XY}	paired C_{XZ}	shared exponent
(λ, λ^{XY})	(r,s)	$V(r,s)$	$U(r,s)$	a_0
$(\lambda^{YZ}, (\lambda^{XY})^{YZ})$	(r,s)	$U(r,s)$	$V(r,s)$	a_0
$(\lambda^{\mathrm{cyc}}, (\lambda^{XY})^{\mathrm{cyc}})$	(s,r)	$U(r,s)$	$U(r,s)$	b_0

For a matched-sector tensor $T = \sum_h T_h$, convert the sector constructions into concavity by applying Corollary 5.2 to λ and to the character obtained by exchanging X, Y, using the same probability vector. Averaging the logarithms gives, with averaged shared-leg exponent $\alpha > 0$,

$$\log G(T) \geq \alpha H(q) + \sum_h q_h \log G(T_h).$$

For positive sector values, put $Z = \sum_h G(T_h)^{1/\alpha}$ and $q_h = G(T_h)^{1/\alpha}/Z$. Then the right side is $\alpha \log Z$, yielding

$$G(T)^{1/\alpha} \geq \sum_h G(T_h)^{1/\alpha}$$

when the averaged exponent of the shared leg is $\alpha > 0$. Apply this argument to each character pair in the table, using degeneration monotonicity for the source. In the determinant construction the source has the extra factor $B_X(2)$, so its paired value is multiplied by 2^{α}. For a character pair (ϕ, ψ), its paired convolution profile is $J(r,s) = \sqrt{\phi(C_{XY}(r,s))\psi(C_{XY}(r,s))}$, with α the average of their X-dot exponents. Consequently,

$$J(r,s+1)^{1/\alpha} + J(r,s-1)^{1/\alpha} \leq 2J(r,s)^{1/\alpha} \quad (s \geq 2).$$

In the third row, $J(r,s) = U(s,r)$, so this is first-coordinate concavity for U^{1/b_0}. In the three-sector construction there is no extra factor: the table gives the two coupled tripling inequalities with exponent a_0, and the self-tripling inequality with exponent b_0. Symmetry supplies the other coordinate of V.

7 The discrete parameter cap

Lemma 7.1. *For every character, its dot parameters satisfy*

$$b_0 \le 4a_0(1-a_0), \qquad a_0 = (p_X + p_Y)/2, \quad b_0 = p_Z. \tag{4}$$

Proof. Fix a character and its profiles just defined. If $a_0 = 0$, the endpoint with $r = 2$ gives $V(2,2) \le 1$, so $2^{b_0} \le V(2,2)$ forces $b_0 = 0$. The case $b_0 = 0$ is immediate since $0 \le a_0 \le 1$. Hence assume $a_0, b_0 > 0$, and put

$$p = 1/a_0 \ge 1, \qquad \theta = b_0/a_0 > 0, \qquad P = V^{1/a_0}, \quad Q = U^{1/a_0}.$$

The remaining goal is $\theta \le 4(p-1)/p$. The determinant construction and its leg permutations make P separately discretely concave, Q concave in its second coordinate, and $Q^{1/\theta}$ concave in its first. Here discrete concavity means that consecutive increments are nonincreasing. Integer coordinate restrictions and Lemma 4.1 give positivity, coordinate monotonicity, symmetry of P, and

$$P(1,s) = s, \quad Q(n,1) = n^\theta, \qquad P(n,s), Q(n,s) \le (n+s-1)^p \quad (n,s \ge 1).$$

The three-sector construction gives

$$\begin{aligned} P(n, 3s+n-1) &\ge 2P(n,s) + Q(n,s), \\ Q(n, 3s+n-1) &\ge 2Q(n,s) + P(n,s), \\ Q(3n+s-1, s)^{1/\theta} &\ge 3Q(n,s)^{1/\theta}. \end{aligned}$$

For $n \ge 2$, the secant inequality for the concave row $P(n,\cdot)$ gives

$$P(n, 4n-1) \le P(n,n) + \frac{3n-1}{n-1}(P(n,n) - P(n,1)).$$

The first tripling inequality and positivity therefore give $Q(n,n) \le \frac{2n}{n-1} P(n,n) \le 4P(n,n)$. A positive nondecreasing concave row satisfies $f(n) \le nf(2)$: for $n \ge 2$, use $f(n) \le f(1) + (n-1)(f(2) - f(1))$, and for $n = 1$ use monotonicity. Apply this in each coordinate of P to obtain

$$n^\theta = Q(n,1) \le Q(n,n) \le 4P(n,n) \le 4P(2,2)n^2.$$

Thus $\theta \le 2$. If $p \ge 2$, this already gives $\theta \le 4(p-1)/p$. It remains to handle $1 \le p < 2$.

Common slopes. Row concavity gives limiting increments and the same linear growth rates: the nonnegative increments decrease, and telescoping followed by Cesàro averaging identifies their limit with $f(s)/s$. For a fixed positive integer n, let L_P, L_Q be these two row limits. Dividing the first two tripling inequalities by s gives $3L_P \ge 2L_P + L_Q$ and $3L_Q \ge 2L_Q + L_P$, respectively, so the two row growth rates agree. Write

$$L_n = \lim_{s\to\infty} \frac{P(n,s)}{s} = \lim_{s\to\infty} \frac{Q(n,s)}{s}.$$

Every row increment is at least L_n, and monotonicity in n makes L_n nondecreasing. Put $H(h) = P(n,h) + Q(n,h)$ for this paragraph. Then $H(3h+n-1) \ge 3H(h)$. Its iterates are $h_j = 3^j(h + (n-1)/2) - (n-1)/2$, so $H(h) \le H(h_j)/3^j$. Since $H(h_j)/h_j \to 2L_n$, this yields

$$P(n,h) + Q(n,h) \le (2h+n-1)L_n \quad (n,h \ge 1).$$

Positivity at $h = 1$ implies $L_n > 0$. With $D_n = P(n, n)$ and $h = \max(n, m)$, use $P(n, h) \geq D_n + (h - n)L_n$ and $Q(n, h) \geq Q(n, m) + (h - m)L_n$ in the preceding bound to obtain

$$Q(n, m) \leq L_n(m + 2n - 1) - D_n.$$

At $m = n$, $Q(n, n) \geq (n - 1)L_n$ gives $D_n \leq 2nL_n$. Also $D_n \geq n + (n - 1)L_n$, since $P(n, 1) = n$. For $n \geq 2$, these imply

$$\tfrac{1}{2} \leq \frac{nL_n}{D_n} \leq 2 \quad (n \geq 2).$$

Eliminate the varying slope. Fix positive integers u, v, let $\tau = u/v$, and, for integers $t \geq 2$, set

$$D(t) = P(vt, vt), \quad W(t) = Q(vt, ut), \quad \ell_t = \frac{vtL_{vt}}{D(t)}, \qquad z = \frac{u - 1}{2v + u} \geq 0.$$

For positive integers n, m, the two-coordinate gains follow in a definite order. First, symmetry and the nondecreasing slopes give $P(n + v, n + v) \geq D_n + vL_n + vL_{n+v} \geq D_n + 2vL_n$. For Q, move in the second coordinate first. The affine bound gives

$$Q(n, m + u) \geq Q(n, m) + uL_n \geq Q(n, m)\left(1 + \frac{uL_n}{L_n(m + 2n) - D_n}\right).$$

The denominator is positive because it strictly exceeds $Q(n, m)$. At fixed second coordinate s, put $f(j) = Q(j, s)^{1/\theta}$. Concavity and self-tripling give, for $1 \leq h \leq 2n + s - 1$,

$$f(n + h) - f(n) \geq \frac{h}{2n + s - 1}(f(3n + s - 1) - f(n)) \geq \frac{2h}{2n + s - 1}f(n).$$

Use $h = v$, $n = vt$, $m = ut$, and then $s = m + u$. The resulting steps are

$$\begin{aligned} D(t + 1) &\geq D(t)(1 + 2\ell_t/t), \\ W(t + 1) &\geq W(t)(1 + F_\tau(\ell_t)/t)(1 + \gamma/(t + z))^\theta, \\ F_\tau(x) &= \frac{\tau x}{(\tau + 2)x - 1}, \qquad \gamma = \frac{2}{2 + \tau}. \end{aligned}$$

The function F_τ is used on $x \geq 1/2$, where its denominator is positive. The varying $\ell_t \in [1/2, 2]$ is the obstacle to summing the second step directly. To eliminate it, put $y = p/2$. The identity

$$F_\tau(x) - F_\tau(y) + \frac{\tau(x - y)}{((\tau + 2)y - 1)^2} = \frac{\tau(\tau + 2)(x - y)^2}{((\tau + 2)x - 1)((\tau + 2)y - 1)^2} \geq 0$$

for $x, y \geq 1/2$ proves the tangent bound

$$F_\tau(x) \geq F_\tau(p/2) - c(x - p/2), \qquad c = \frac{\tau}{((\tau + 2)p/2 - 1)^2} \geq 0 \quad (x \geq 1/2).$$

The tangent contributes $-c\ell_t/t$, which is cancelled by $(c/2)$ times the diagonal logarithmic increment. Accordingly, the potential $\Phi(t) = \log W(t) + (c/2)\log D(t)$ satisfies

$$\begin{aligned} \Phi(t + 1) - \Phi(t) &\geq \frac{\theta\gamma + F_\tau(p/2) + cp/2}{t} - O(t^{-2}), \\ \Phi(t) &\leq p(1 + c/2)\log t + O(1). \end{aligned}$$

The first estimate uses $\log(1+x) \geq x - x^2/2$; the bounded slopes make the error uniform in t for this fixed ray. The second uses the rank bounds. Summing telescopically, the t^{-2} errors remain bounded and the $cp/2$ terms cancel, leaving

$$\frac{2\theta}{2+\tau} + \frac{\tau}{\tau+2-2/p} \leq p.$$

This holds for every positive rational τ, hence for every $\tau > 0$ by continuity. Taking $\tau = 4/p - 2 > 0$ gives $\theta \leq 4(p-1)/p$. Since $b_0 = a_0\theta$ and $p = 1/a_0$, this is (4). □

8 The finite scheme and its arithmetic cost

Proof of Theorem 1.1. Fix $k \geq 0$ and $\varepsilon > 0$, and put $\eta = \varepsilon/2$. Choose an integer $q \geq 2$ with $2 < q^\eta$, and let $j = \lceil q^k \rceil$. For every character, $j \leq 2q^k$ and $0 \leq b_0 \leq 1$, so (3) and Lemma 7.1 give

$$\lambda(\langle q, j, q\rangle) \leq 2q^{2a_0 + kb_0} \leq 2q^{\max_{0\leq x\leq 1}(2x+4kx(1-x))} = 2q^{f(k)}.$$

The quadratic maximum occurs at $x = 1$ for $k \leq 1/2$, and at $x = 1/2 + 1/(4k)$ otherwise. Hence (1) gives

$$\widetilde{\operatorname{rank}}_{\mathbb{Z}}(\langle q, j, q\rangle) \leq 2q^{f(k)} < q^{f(k)+\eta}.$$

There is therefore a fixed integer $m \geq 1$ with an exact integer rank decomposition of length $\rho < q^{m(f(k)+\eta)}$. Reindexing this tensor power gives $\langle a, b, a\rangle$, where $a = q^m$, $b = j^m$. Its three flattenings over $\mathbb{Q}$ supply the last condition in the finite base certificate

$$a \geq 2, \qquad b \geq a^k, \qquad \log_a \rho < f(k) + \eta, \qquad \rho \geq \max\{ab, a^2\}. \tag{5}$$

Only this fixed scheme is needed from now on.

Correctness over every ring. Let R be any associative ring, and let $A \in R^{a\times b}$ and $B \in R^{b\times a}$. Writing the chosen integer decomposition coefficients as α, β, γ gives, for $1 \leq i, h \leq a$,

$$(AB)_{ih} = \sum_{\ell=1}^{\rho} \gamma_{ih,\ell} \left(\sum_{i',j'} \alpha_{\ell,i'j'} A_{i'j'} \right) \left(\sum_{j'',h'} \beta_{\ell,j''h'} B_{j''h'} \right),$$

where the entry sums run over the indicated matrix coordinates. Expansion leaves each A-entry to the left of its B-entry, and integer coefficient equality proves the identity. Integer multiples use only addition and negation. The same expansion applies to rectangular block multiplication, a $\mathbb{Z}$-bilinear map, so it also justifies the recursive scheme.

Arithmetic cost and padding. An integer coefficient c uses at most $|c| + 1$ gates, and adding its term to an accumulator uses one further gate. Let K_0 be the sum of $|c| + 2$ over all entries of the three fixed coefficient arrays, and set $d = \max\{ab, a^2\} \leq \rho$. The preparation and reconstruction at depth $t + 1$ act on blocks with $(ab)^t, (ab)^t, (a^2)^t$ entries. Each of the ρ recursive products is computed once and reused in reconstruction. Thus the actual gate count $T(t)$ satisfies

$$T(0) = 1, \qquad T(t+1) \leq \rho T(t) + K_0 d^t.$$

Induction using $d \leq \rho$ gives

$$\operatorname{cost}(t) = T(t) \leq (1 + K_0)(t+1)\rho^t.$$

For an integer $n \geq 1$, choose $t = \lfloor \log_a n \rfloor + 1$. Then

$$n < a^t \leq an, \qquad \lceil n^k \rceil \leq b^t.$$

Zero input padding and output wires give the requested shape. The inner padding need not have a constant ratio to n^k; its operations are already included in $\mathrm{cost}(t)$. Since $\log_a \rho \geq 0$ and $t + 1 = O(1 + \log n)$,

$$\mathrm{cost}(t) = O\big((1 + \log n) n^{\log_a \rho}\big) = O_{k,\varepsilon}(n^{f(k)+\varepsilon}).$$

Here $1 + \log n = O_\eta(n^\eta)$ and (5) reserves the other η of slack. A fixed constant covers every $n \geq 1$. The base dimensions, coefficients, and operation counts depend only on k, ε, so this constant is independent of R. □

9 Supplementary exponent bounds

For nonnegative shape exponents u, v, w, let $\omega_{\mathrm{univ}}(u, v, w)$ be the infimum of arithmetic exponents of universal integer bilinear circuits for shapes $\lceil n^u \rceil \times \lceil n^v \rceil$ and $\lceil n^v \rceil \times \lceil n^w \rceil$. Let $\omega_R(u, v, w)$ be its counterpart when algorithms may depend on the fixed ring R. The main theorem gives

$$\omega_R(1, k, 1) \leq \omega_{\mathrm{univ}}(1, k, 1) \leq f(k).$$

For the universal lower bound, evaluate the integer coefficient identities over $\mathbb{Q}$. Output flattening gives at least n^2 products, hence $\omega_{\mathrm{univ}}(1, k, 1) = 2$ for $0 \leq k \leq 1/2$. In particular the universal dual exponent is at least $1/2$. A fixed degenerate ring can have weaker arithmetic lower bounds; counting explicit dense output writes also gives exponent two for that output model.

Corollary 9.1 (Three arbitrary dimensions)**.** *Let $a \geq b \geq c \geq 0$ be the decreasing rearrangement of three shape exponents. Uniformly over all associative rings,*

$$\omega_R(a, b, c) \leq \omega_{\mathrm{univ}}(a, b, c) \leq \begin{cases} a + b, & b \geq 2c, \\ a + b + \dfrac{(2c - b)^2}{4(a + c - b)}, & b < 2c. \end{cases}$$

The bound holds for every permutation of the shape exponents, and the universal exponent equals $a + b$ when $b \geq 2c$.

Proof. Permute (4) to obtain the three symmetric parameter constraints. These permutations act on integer tensor identities; each resulting scheme still places the first input on the left and the second on the right. For any character, let (p_1, p_2, p_3) be the permutation of its dot parameters corresponding to the sorted shape weights, and put $S = p_1 + p_2 + p_3$. If $S \leq 2$, then $p_i \in [0, 1]$ bounds the objective $ap_1 + bp_2 + cp_3$ by $a + b$. If $S > 2$, the three cap constraints imply $S \leq 9/4$ and, with $d = \sqrt{9 - 4S} \in [0, 1]$,

$$\frac{3 - 2d - d^2}{4} \leq p_i \leq \frac{3 + 2d - d^2}{4}.$$

Using these endpoints and $p_1 + p_2 + p_3 = S$ bounds the objective by

$$\frac{3(a + b + c)}{4} + \frac{a - c}{2} d - \frac{a + c - b}{4} d^2.$$

Its maximum on $[0, 1]$ is the displayed bound; in the second branch, $b < 2c$ ensures $a + c - b > 0$.

For rational shape exponents, choose a common integer base $A \geq 2$ so A^a, A^b, A^c are integers. The character bound and spectral duality produce a finite power scheme with rank ratio below the claimed bound plus any positive slack. At common depth $t = \lfloor \log_{A^m} n \rfloor + 1$, where $m \geq 1$ is that fixed power, the same level estimate gives $O((t+1)\rho^t)$ operations. For real exponents, approximate the sorted triple from above by rational triples and use continuity of the bound and padding. When all exponents are zero, a single scalar product suffices. Finally, the three flattenings over $\mathbb{Q}$ give the universal lower bound $\max\{a+b, a+c, b+c\} = a+b$. □

Verification. The accompanying `verify_integral_constructions.py` checks finite instances of the separator, determinant bases and blocks, and the oriented tripling identity using exact integer arithmetic. These checks supplement the general proofs. The `formalization` directory proves Theorem 1.1, including coefficient arithmetic, with strict addition, negation, and ordered multiplication syntax. The witnesses are classical circuit existence proofs; no executable implementation or evaluator runtime bound is extracted. The three-dimension corollary and lower bounds are supplementary mathematical statements outside the public Lean theorem.

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